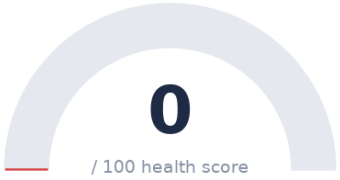
Dataset Audit Report

sales_transaction.csv | target: **Return_Flag** | 07 Oct 2026



Verdict: Not ready: fix the critical issues before training

6000 rows, **232** features (after encoding), **2** classes. The audit raised **3 critical** and **11 warning** finding(s). Model-based checks used a stratified sample of 6,000 of 18,045 rows; the duplicate rate was measured on all rows.

MLP training budget: about **2 epochs** (batch 64, ~142 optimizer steps), roughly **0.2 s** on this machine. Best validation epoch was 1; recommendation adds a 20% margin.

Risk assessment

Risk	Level	Why
Overfitting	HIGH	train/validation gap 0.50, noisy labels get memorised, leak columns inflate scores
Underfitting	HIGH	no signal over shuffled labels, train and validation both low
Shortcut learning	LOW	ID-like column present
Data quality	HIGH	8 hygiene issue(s) flagged

Findings

Level	Area	Finding	What to do
FAIL	Missing data	'Promotion_Code' has 54% missing values	Impute inside a pipeline (fit on train only) or drop the column if >50% empty.
FAIL	Signal	Real-label score (0.50) is not clearly above shuffled-label noise (0.50)	Features carry no learnable information about the target. Training will not help; revisit features or the target.
FAIL	Label noise	~46% of labels look wrong (2737 rows confidently contradicted by the model)	Review the listed suspect rows (see appendix). Noisy labels cap accuracy and make models memorise errors.
WARN	Missing data	'Customer_Rating' has 19% missing values	Impute inside a pipeline (fit on train only) or drop the column if >50% empty.
WARN	Shortcut risk	'Transaction_ID' looks like an ID / row index (5997 unique values)	Identifiers let flexible models memorise rows. Drop it.
WARN	Shortcut risk	'Order_ID' looks like an ID / row index (5581 unique values)	Identifiers let flexible models memorise rows. Drop it.
WARN	Shortcut risk	'Order_Time' looks like an ID / row index (5670 unique values)	Identifiers let flexible models memorise rows. Drop it.
WARN	Class balance	Class imbalance 8.6:1	Use stratified splits, class weights or resampling, and report F1 / PR-AUC / balanced accuracy.
WARN	Outliers	'Sales_Amount' has 6% extreme outliers (beyond 3 x IQR)	Check for entry errors; consider clipping / log transform / robust scaling.
WARN	Outliers	'Cost_Amount' has 6% extreme outliers (beyond 3 x IQR)	Check for entry errors; consider clipping / log transform / robust scaling.
WARN	Outliers	'Profit' has 6% extreme outliers (beyond 3 x IQR)	Check for entry errors; consider clipping / log transform / robust scaling.
WARN	Redundancy	101 near-duplicate feature pair(s), e.g. 'Store_ID_AUS-01' & 'Store_Name_Austin Business' (r=1.00)	Drop one of each pair; unstable coefficients and misleading importances otherwise.
WARN	Underfitting	Logistic regression: both train (0.50) and validation scores are low	High bias. Try a more flexible model or better features.

Level	Area	Finding	What to do
WARN	Overfitting	Random forest: train 1.00 vs validation 0.50 (gap 0.50)	High variance. Add regularisation, simplify the model, get more data or reduce features.
INFO	Distributions	5 heavily skewed feature(s): Quantity, Unit_Price, Sales_Amount, Cost_Amount, Profit	Log / power transform helps linear models and MLPs (trees do not care).
INFO	Training	Validation loss bottoms out after only 1 epoch(s)	Easy or tiny problem, or an overly large learning rate. Use early stopping and a lower LR.

1. Data overview and hygiene

Class balance and missingness decide which metrics and preprocessing are valid. Heavy imbalance makes plain accuracy misleading; missing values must be imputed inside the cross-validation loop to avoid leaking statistics from the test fold.

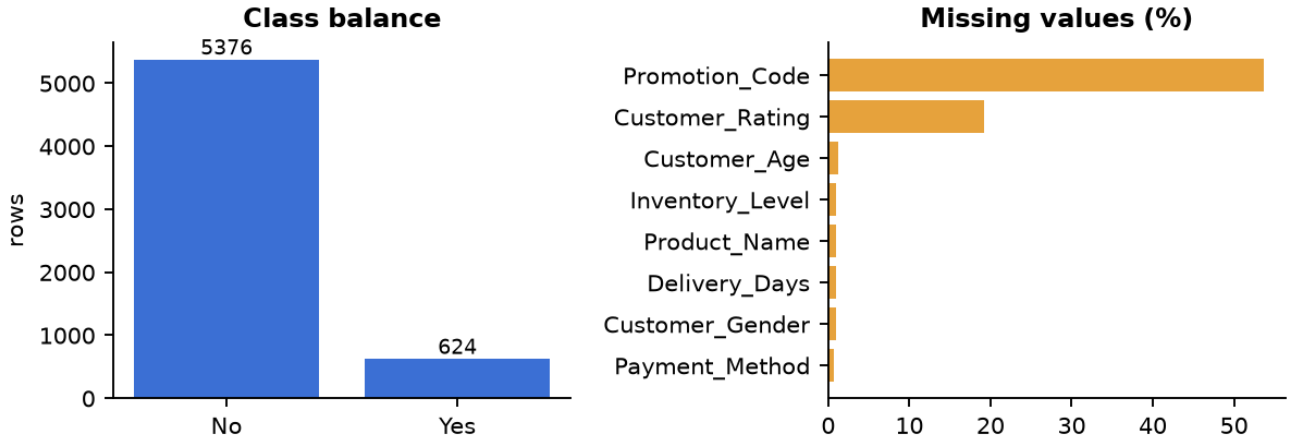


Figure 1. Class distribution and columns with missing values.

Column profile (first 22 columns)

Column	Type	Missing	Unique	Mean	Std	|Skew|	Outliers
Transaction_ID	str	0.0%	5997	-	-	-	-
Order_ID	str	0.0%	5581	-	-	-	-
Customer_ID	str	0.0%	3224	-	-	-	-
Customer_Name	str	0.6%	1116	-	-	-	-
Customer_Age	float64	1.2%	58	38.8	10.9	0.2	0%
Customer_Gender	str	0.9%	6	-	-	-	-
Customer_Segment	str	0.0%	6	-	-	-	-
Order_Date	str	0.0%	1406	-	-	-	-
Order_Time	str	0.0%	5670	-	-	-	-
Sales_Channel	str	0.0%	4	-	-	-	-
Store_ID	str	0.0%	12	-	-	-	-
Store_Name	str	0.0%	12	-	-	-	-
Country	str	0.0%	6	-	-	-	-
Region	str	0.0%	14	-	-	-	-
City	str	0.0%	15	-	-	-	-
Product_ID	str	0.0%	28	-	-	-	-
Product_Name	str	1.0%	28	-	-	-	-
Product_Category	str	0.0%	7	-	-	-	-
Product_Subcategory	str	0.0%	17	-	-	-	-
Quantity	int64	0.0%	12	2.14	1.8	2.6	1%
Unit_Price	float64	0.0%	5137	231	294	2.1	3%
Discount_Percentage	int64	0.0%	7	6.39	6.89	0.8	0%

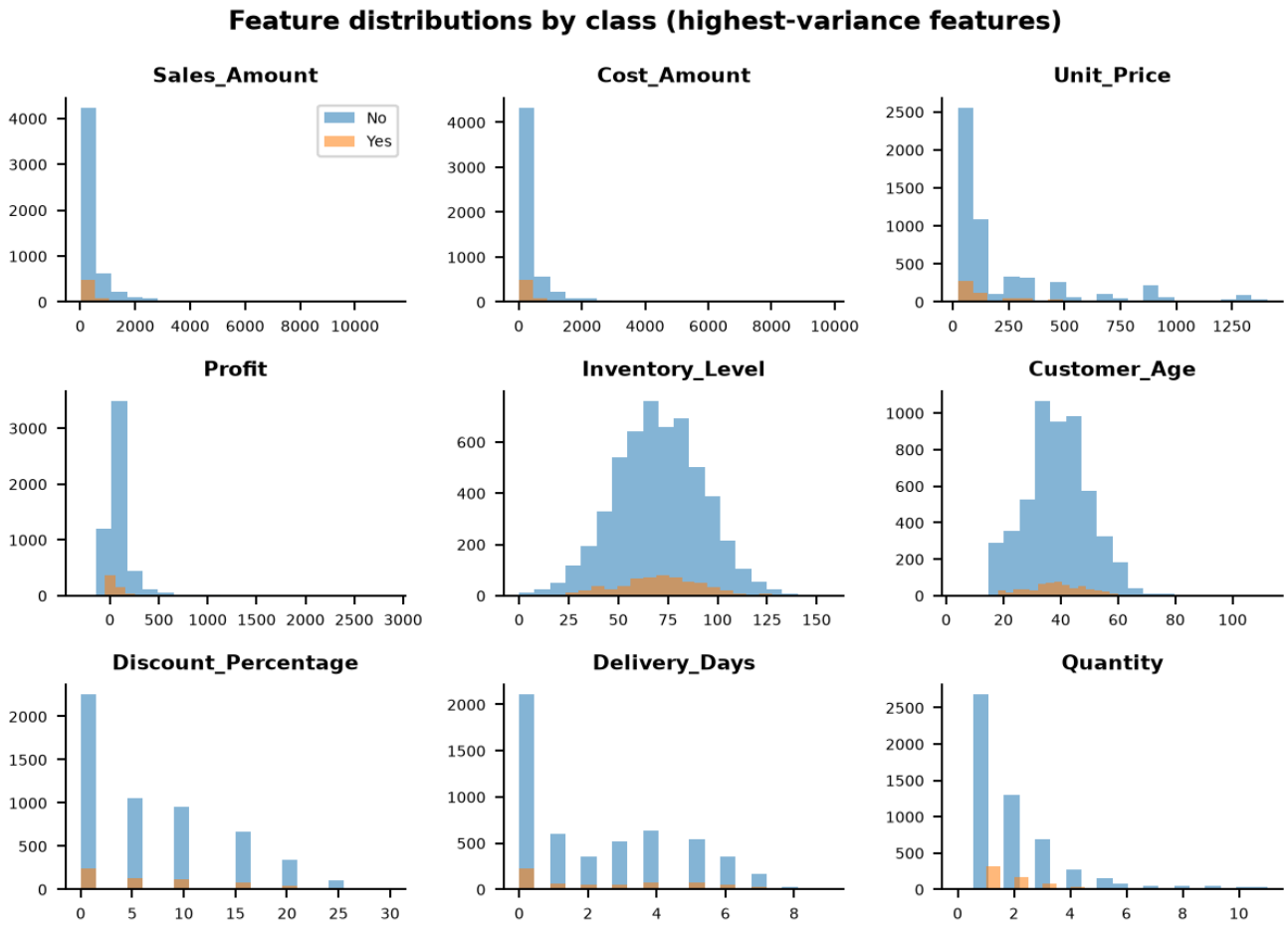


Figure 2. Distributions split by class. Strong separation means an easy feature; identical overlap means weak information.

2. Redundancy and correlation

Highly correlated features carry the same information twice. They destabilise linear-model coefficients, split importance between copies, and add dimensions without adding signal.

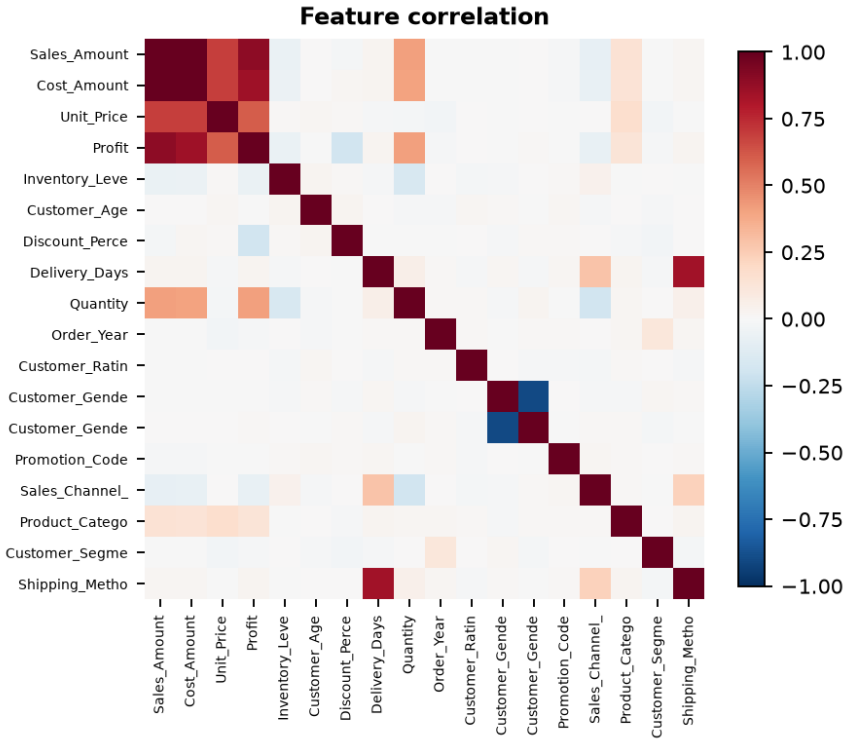


Figure 3. Pairwise correlation of the 18 highest-variance features.

Near-duplicate pairs (|r| > 0.95): 101

Feature A	Feature B	|r|
Store_ID_AUS-01	Store_Name_Austin Business	1.000
Store_ID_AUS-01	Region_South	1.000
Store_ID_AUS-01	City_Austin	1.000
Store_ID_CHI-01	Store_Name_Chicago Loop	1.000
Store_ID_CHI-01	Region_Midwest	1.000
Store_ID_CHI-01	City_Chicago	1.000
Store_ID_COL-01	Store_Name_Cologne Business	1.000
Store_ID_COL-01	City_Cologne	1.000
Store_ID_LA-01	Store_Name_Los Angeles Downtow	1.000
Store_ID_LA-01	Region_West	1.000
Store_ID_LON-01	Store_Name_London City	1.000
Store_ID_LON-01	City_London	1.000

3. Shortcut learning and leakage

A model will happily learn whatever predicts the label fastest, including cues that will not exist in production. Two tests: (a) can any *single* feature predict the label almost perfectly? (b) does one feature account for most of the model's skill? Red bars above the dashed line (0.95) are leakage suspects.

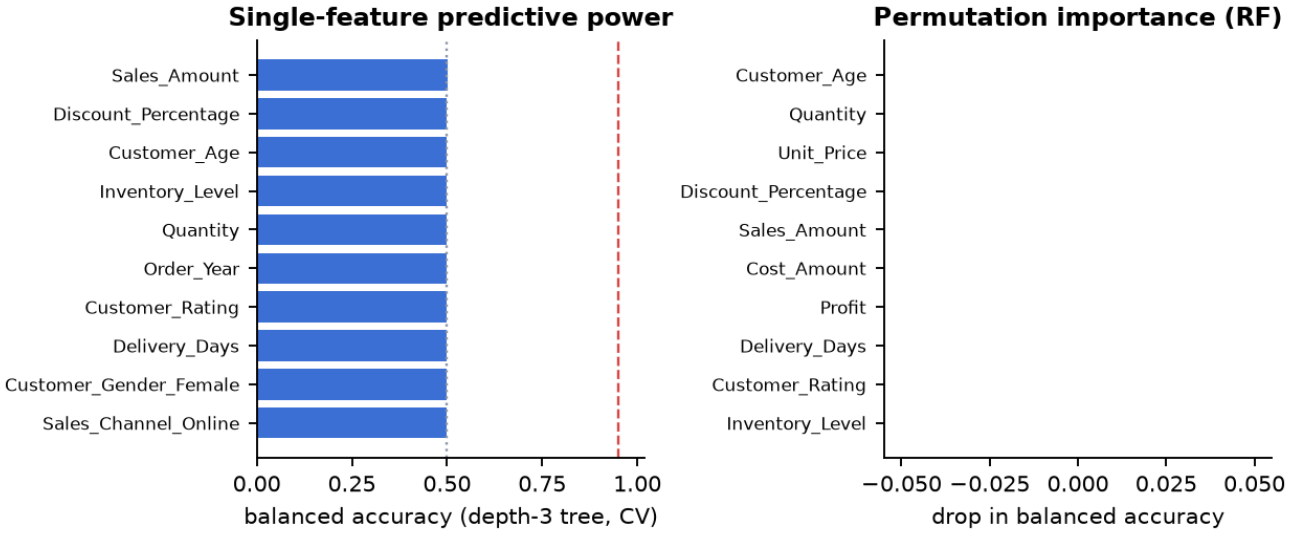


Figure 4. Left: single-feature predictive power. Right: permutation importance of a random forest on held-out rows.

Suspect columns: Transaction_ID, Order_ID, Order_Time. Remove them (or prove they are known at prediction time) and re-run the audit.

Is there learnable signal at all?

We cross-validate a logistic regression on the real labels and on 15 randomly shuffled copies of the labels. If the real score does not clearly beat the shuffled distribution, the features contain no usable information.

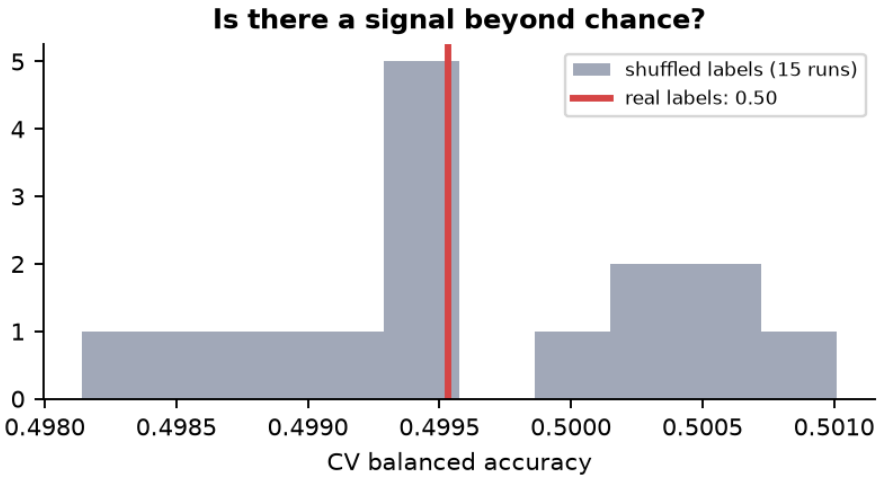


Figure 5. Real-label score against the shuffled-label baseline.

4. Will it overfit or underfit?

Learning curves show how train and validation scores change with data size. A large gap that does not close means high variance (overfitting); two low, converged curves mean high bias (underfitting); a validation curve that is still climbing means more data would help.

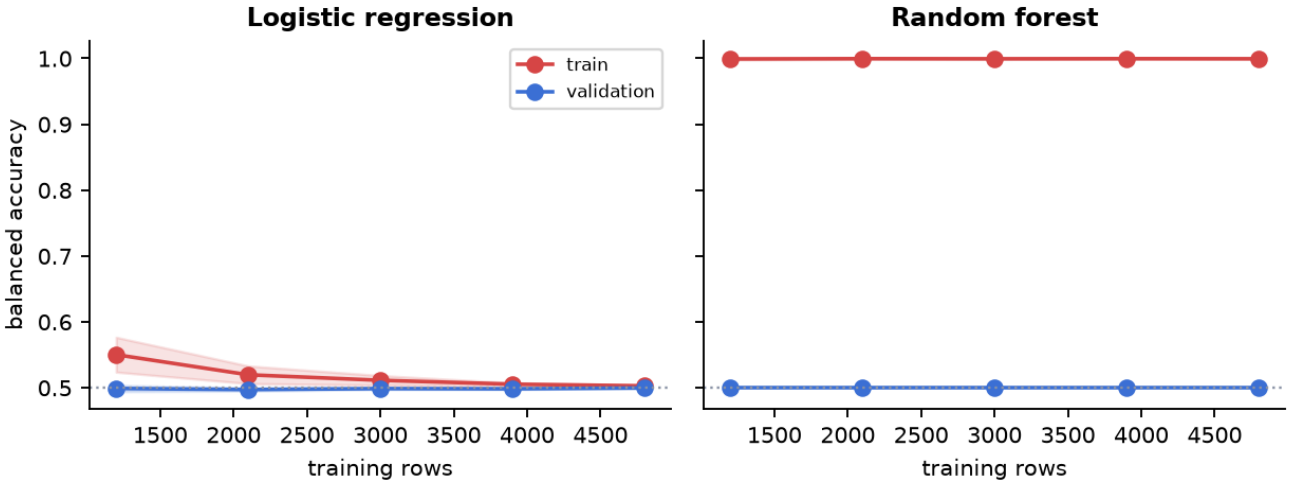


Figure 6. Learning curves (band = 1 std over CV folds; dotted = guessing baseline).

Model-complexity sweep

Increasing tree depth moves the model from underfit (left) to overfit (right). Validation accuracy peaks at depth **9**; deeper trees only memorise the training set.

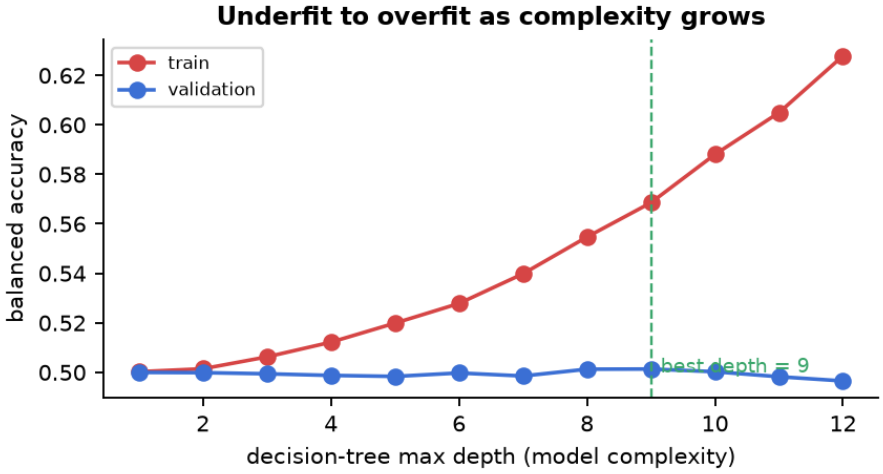


Figure 7. Train vs validation score as complexity increases.

5. Robustness and label quality

A trustworthy dataset yields models that stay accurate under small perturbations and whose labels agree with the evidence in the features.

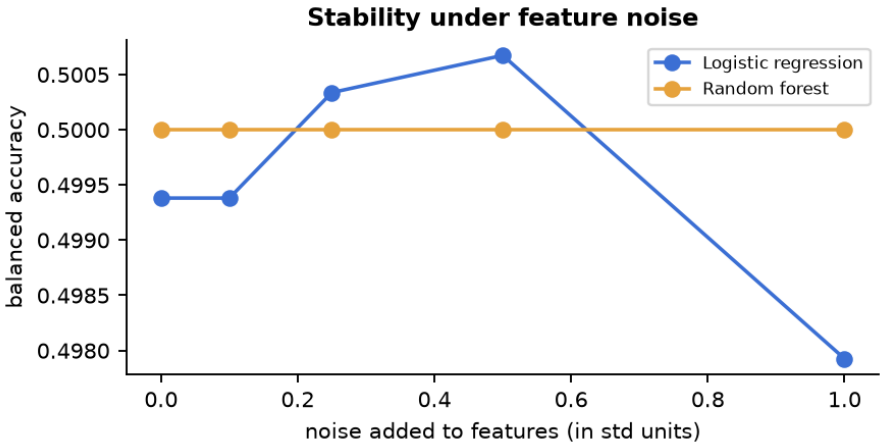


Figure 8. Held-out balanced accuracy when Gaussian noise is added to the (standardised) features.

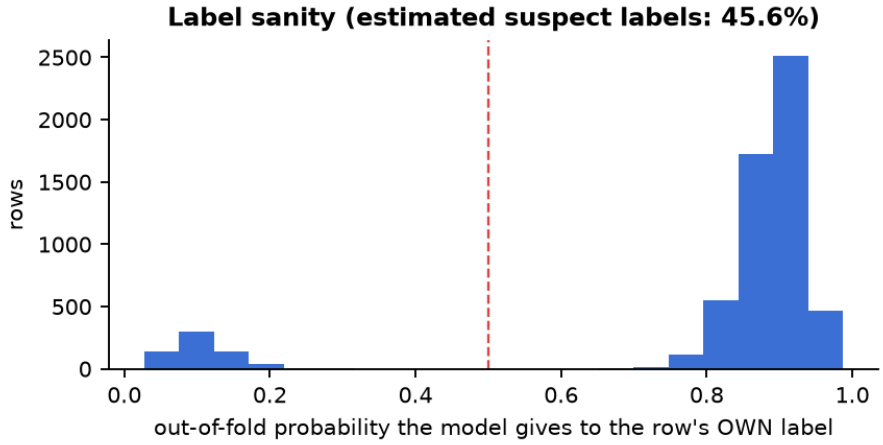


Figure 9. Out-of-fold confidence in each row's own label. A pile-up near 0 indicates mislabeled rows (simplified confident learning).

6. Training budget

A small MLP (2 x 64 hidden units, Adam, batch 64) was trained with early stopping. Validation loss was lowest at epoch **1**. Recommended: **2 epochs** with early stopping (patience about 10-20), about **0.2 s** total here (98 ms per epoch). Treat this as an estimate for this dataset and this machine.

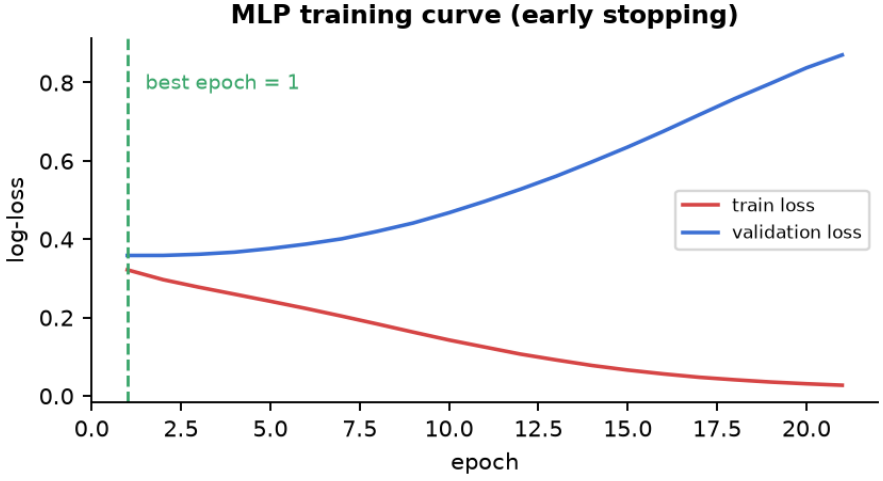


Figure 10. Train and validation loss per epoch; the dashed line marks the minimum validation loss.

7. Recommended actions

1. Impute inside a pipeline (fit on train only) or drop the column if >50% empty.

2. Features carry no learnable information about the target. Training will not help; revisit features or the target.

3. Review the listed suspect rows (see appendix). Noisy labels cap accuracy and make models memorise errors.

4. Identifiers let flexible models memorise rows. Drop it.

5. Use stratified splits, class weights or resampling, and report F1 / PR-AUC / balanced accuracy.

6. Check for entry errors; consider clipping / log transform / robust scaling.

7. Drop one of each pair; unstable coefficients and misleading importances otherwise.

8. High bias. Try a more flexible model or better features.

9. High variance. Add regularisation, simplify the model, get more data or reduce features.

10. Split with stratification and fit every preprocessing step (imputer, scaler, encoder) inside a Pipeline on the training fold only.

11. Report balanced accuracy / F1 / ROC-AUC alongside accuracy, using repeated stratified k-fold on small data.

12. Start with simple models (logistic regression, shallow trees up to depth 9) as the baseline before trying MLPs.

Methodology and limits

Checks: IQR outliers (3 x IQR), Pearson correlation, single-feature depth-3 trees (5-fold CV), permutation importance, label-shuffle test (15 shuffles), learning curves and depth sweep with stratified CV, Gaussian-noise stability test, simplified confident-learning label audit, early-stopped MLP. The health score is a heuristic: 15 points per critical finding, 5 per warning, 25-30 for leakage or no signal, capped per area. All estimates come from this dataset only; small datasets give wide error bars. This tool targets tabular classification.

Appendix: suspect label rows

Row indices (0-based, after dropping rows with a missing target) whose label is confidently contradicted by the model. Inspect them manually before deciding to relabel or remove.

Row	Given label	Model's own-label prob
3337	Yes	0.03
634	Yes	0.03
4197	Yes	0.03
4043	Yes	0.03
3314	Yes	0.04
5991	Yes	0.04
4081	Yes	0.04
5469	Yes	0.04
1815	Yes	0.04
659	Yes	0.04
3240	Yes	0.04
5175	Yes	0.04
432	Yes	0.04
3710	Yes	0.04
704	Yes	0.04
4834	Yes	0.05
2543	Yes	0.05
953	Yes	0.05
1314	Yes	0.05
433	Yes	0.05
2248	Yes	0.05
6	Yes	0.05
290	Yes	0.05
2363	Yes	0.05
550	Yes	0.05
1374	Yes	0.05
4420	Yes	0.05
4055	Yes	0.05
4012	Yes	0.05
2270	Yes	0.05
1784	Yes	0.05
945	Yes	0.05
2750	Yes	0.05
547	Yes	0.05
207	Yes	0.05
4981	Yes	0.05

Row	Given label	Model's own-label prob
2266	Yes	0.06
296	Yes	0.06
2111	Yes	0.06
4817	Yes	0.06